Fire Without Forgetting: Persistent Dendrites Give Spiking World Models Long-Term Memory

Anonymous CVPR submission

Paper ID *****

Abstract

Embodied agents see only part of their world and must remember what has left their view. World models give an agent this memory through a recurrent latent state, and spiking neural networks (SNNs) promise to run them with sparse binary events on low-power hardware. We find that spiking world models forget within a few steps, although their neurons have long time constants. Membrane potentials are cleared at every environment step and dendrites are reset by every somatic spike, so memory has to be re-encoded into binary spikes at each step and the time constants have no effect. Causal interventions confirm this reset-induced amnesia*: a spiking world model fails to recall a cue 8 steps back whatever its time constants, whereas with both resets removed and its write gain normalized it recalls cues 32 steps back. We propose FwF, a spiking world model in which dendrites store and spikes communicate. Each neuron keeps linear dendritic channels that are never reset and never receive spikes back, with retention times initialized from 1 to 1000 environment steps and a write gain normalized across them. Half of the channels resonate, so that their phase records how long ago an input arrived. A stateless soma reads the channels and emits a binary spike, the only signal that neurons of the world model exchange. On BSuite memory length, FwF is the only model we train, spiking or not, that reaches a return of 0.9 at 64 steps within 500k environment steps; with the lengths and budget of R2I it also does so at 71 and 101 steps, where DreamerV3 stays near chance. Its world model is the only one that recalls the input of exactly 32 steps ago on POPGym RepeatPrevious, and the agent uses 4.3× less arithmetic energy per step than DreamerV3.*

1. Introduction

An agent that explores a house through a head-mounted camera sees a few square meters at a time. To return to the kitchen, or to pick up the object it passed a minute ago, it must remember what it can no longer see. World models provide this memory: they compress the history of observations and actions into a recurrent latent state, predict the future from it, and let the agent learn behavior in imagination [10–12, 14]. Recent world models remember for hundreds of steps [7, 37], and memory benchmarks measure how far [25, 31, 32].

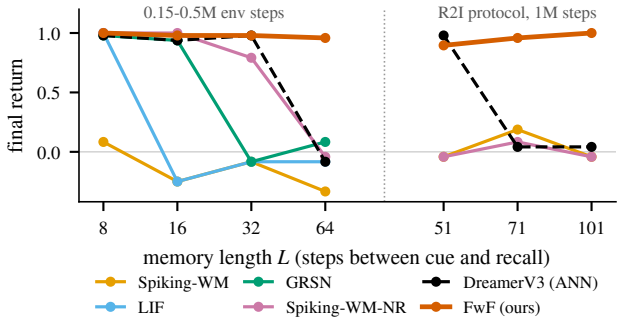


Figure 1. **Spiking world models forget.** Final return on BSuite memory length [31]: the agent sees a cue, then L uninformative steps, and is rewarded for recalling the cue (1 = always correct, 0 = chance). All models share one spiking world model and differ in the recurrent core (Sec. 5.1). Left: 0.15–0.5M environment steps. Right: $L = 51$–101 with the lengths and budget of R2I [37] (1M steps). Spiking baselines reach a return of 0.9 up to 32 steps and DreamerV3 (ANN) up to 51; FwF is the only model that reaches it beyond 51 steps (Tab. 2).

Embodied agents also run on batteries. Spiking neural networks (SNNs) communicate with binary events, replace multiply-accumulates by accumulates, and run on neuromorphic processors at a fraction of the energy of dense networks [6, 29, 36]. Spiking neurons are dynamical systems with membrane and dendritic time constants, and these dynamics are widely credited with a capacity for temporal memory [3, 49, 51]. Spiking-WM [41] builds a spiking world model on multi-compartment neurons whose dendrites are meant to integrate information over long timescales.

We find that spiking world models do not use this capacity: they forget within a few steps (Fig. 1). On BSuite

memory length, Spiking-WM does not recall a cue 8 steps back; recurrent LIF neurons stop at 8 steps and gated recurrent spiking neurons [34] at 16. The cause is reset. First, every environment step starts from cleared membrane potentials, the standard practice in spiking RL [21, 41, 42]; only the binary spike output reaches the next step. Second, a somatic spike resets the dendrites together with the soma, so a dendritic trace survives only until the neuron next fires. Memory must therefore be re-encoded into spikes at every step, gradients cross a surrogate derivative at every step, and the time constants shape only the few simulation steps inside one environment step. Long time constants do not help even without the resets: the Euler update writes input with gain $1/\tau$, so a compartment with a time constant of 128 simulation steps stores a cue at about one percent of the firing threshold.

We call this failure *reset-induced amnesia* and test it causally (Sec. 3). In a world-model probe that isolates memory from exploration, removing the dendritic reset or restoring the write gain has no effect while the step reset remains. With persistent state the world model recalls cues 16 steps back, and adding the write gain extends this to 32 steps, learned faster than with a GRU core. Conversely, clearing the dendrites of our own model at every step erases its memory.

These results give a design rule: memory should live in a state that no spike ever resets, and spikes should only carry information out of it. We build FwF from *persistent dendritic neurons* (PDNs) that follow this rule (Sec. 4). A PDN keeps linear dendritic channels across environment steps that are never reset, receive no spikes back from the soma, and are written with a gain normalized across time constants from 1 to 1000 environment steps. Integrator channels hold *what* was seen; resonant channels, damped oscillators in the tradition of resonate-and-fire neurons [19], also encode *when*. A stateless soma reads the dendrites and emits one binary spike per step, the only signal that other neurons of the world model receive; the graded state stays inside the neuron, as in neuromorphic cores with multi-variable neurons [29].

We evaluate on memory benchmarks with published ANN protocols and on a non-memory control (Sec. 5). Of all models we train, spiking or ANN, FwF is the only one that reaches a return of 0.9 on BSuite memory length 64 within 500k steps, and with the lengths and budget of R2I [37] it also does so at 71 and 101 steps, where DreamerV3 stays near chance. Its world model is the only one that recalls the card of exactly 32 steps ago in POPGym RepeatPrevious [25] (AUC 0.93; all others, DreamerV3 included, stay at chance). The agent uses 4.3× less arithmetic energy per step than DreamerV3. Our contributions are:

- We identify reset-induced amnesia in spiking world models (step reset, dendritic reset, a write gain tied to the time constant) and establish it with causal interventions.
- We propose the persistent dendritic neuron, in which persistent, normalized integrator and resonant dendrites store memory and a stateless soma communicates by spikes, and build the spiking world model FwF from it.
- We give the first evaluation of spiking world models on standard memory benchmarks (BSuite memory length up to 101 steps, POPGym RepeatPrevious, Memory Maze), with DMC as a non-memory control and per-step energy accounting.

2. Related Work

World models and memory. Latent world models learn a recurrent state from observations and train behavior in imagination [10–14]. Their memory is that of the recurrent core: a GRU in the Dreamer family, attention over a context window in transformer world models [24, 35, 50], and a state-space recurrence in S4WM [7] and R2I [37]. R2I solves memory tasks that DreamerV3 cannot, and we use its settings for BSuite memory length [31] and POPGym [25]. Memory Maze [32] measures long-term memory in first-person 3D navigation with an offline probe of the world-model state, in the spirit of probing map-like representations in navigation agents [45]. Ni *et al*. [28] separate memory length from credit-assignment length; our world-model probes do the same by removing the policy. All these models are dense ANNs.

Memory cells for RL. Dense memory cells built on linear recurrences with decaying, rotating state include the LRU [30], Legendre memory units [44], S5 layers for in-context RL [23], and fast and forgetful memory, whose complex traces drift in phase with time [26]. They pass real-valued vectors between units; a PDN keeps such state inside each spiking neuron, without gates, and sends only binary spikes.

Spiking reinforcement learning. Spiking actors and Q-networks [21, 42] simulate a feedforward SNN for a few steps per environment step and clear it before the next. Spiking-WM [41] brings multi-compartment neurons into a DreamerV3-style world model (DeepMind Control Suite from pixels, speech classification). GRSN [34] aligns a gated recurrent spiking neuron with environment time and carries its state across steps for model-free RL in POMDPs. Spiking model-based control has also been trained end to end with recurrent LIF networks [17]. None of these works evaluates long-horizon memory; we show that the reset conventions they inherit cap memory at a few to a few tens of steps.

Spiking neurons with long memory. Adaptive thresholds [3, 47], two-compartment neurons [49], heterogeneous

dendritic time constants [51] and resonate-and-fire dynamics [15, 19, 48] extend the memory of spiking neurons. Removing the reset makes the membrane linear and parallelizable [9], and spiking state-space layers attach spiking outputs to linear recurrences [2, 8, 38, 40]. These neurons are trained and tested on offline sequence classification, where the network sees the whole sequence and is reset between samples. A world model instead filters an unbounded stream, step by step, and its state is also rolled forward in imagination. FwF is designed for this setting: memory is carried across environment steps only in dendrites that no spike touches, retention times are set in environment steps, and the soma that produces spikes holds no state at all. Biological dendrites act as computational subunits [4, 20, 22, 33] and many neurons resonate below threshold [18]; PDNs borrow both properties in functional form.

3. Reset-Induced Amnesia

3.1. Spiking world models

We follow the recurrent state-space model of DreamerV3 [14]. At environment step t a recurrent core updates a deterministic state h_t from h_{t-1}, the previous stochastic latent z_{t-1} and action a_{t-1}; the posterior infers z_t from h_t and the observation embedding, the prior predicts it from h_t alone, and decoder, reward and continuation heads read (h_t, z_t). In a spiking world model [41] every layer is spiking and is simulated for T steps per environment step ($T = 5$), and $h_t \in \{0,1\}^{T\times N}$ is the spike output of the core.

Spiking-WM uses multi-compartment neurons (MCNs). In simulation step k of environment step t, a basal and an apical dendrite integrate synaptic input with Euler-discretized leaks, and the soma integrates the basal potential gated by the apical one:

$$v^c[k] = v^c[k-1] + \tfrac{1}{\tau_c}\big(I^c[k] - v^c[k-1]\big), \quad c \in \{\mathrm{bas}, \mathrm{api}\},$$
$$s[k] = \mathrm{H}\big(v^{\mathrm{som}}[k] - \vartheta\big), \tag{1}$$

where $I^c[k] = W_c\,[\,x_t[k];\, h_{t-1}[k]\,]$ combines the current input spikes with the core's spikes of the previous environment step and H is the Heaviside step, trained with a surrogate derivative [27, 46]. Two resets act on this neuron:

1. **Step reset.** All compartments are set to zero at the start of every environment step.
2. **Spike reset.** When the soma fires, the soma and both dendrites are set to zero.[1]

Consequences. With the step reset, the core computes $h_t = \Phi(x_t, h_{t-1})$ from a zero state: it is a binary recurrent network, and everything it knows about the past must be re-encoded into the $T \times N$ bits of h_{t-1} at every step. The gradient from a reward L steps after a cue crosses L surrogate derivatives. The time constants τ_c act only within the T simulation steps of one environment step, which explains why changing them barely changes the performance of Spiking-WM [41]. The spike reset additionally truncates a dendritic trace at the next spike: with $\alpha = 1 - 1/\tau_c$,

$$\frac{\partial v^c[k]}{\partial I^c[k-j]} = \frac{1}{\tau_c}\,\alpha^j \prod_{i=1}^{j}\big(1 - s[k-i]\big), \tag{2}$$

which vanishes as soon as the neuron has fired once, so the effective memory is the shorter of τ_c and the inter-spike interval.

[1] The equations of [41] show the somatic reset; the dendritic reset is part of its official implementation (`MCNode.calc_spike`), which all our Spiking-WM runs use.

Table 1. **Interventions on Spiking-WM** (world-model probe, memlen-L-recall). Updates until the reward AUC exceeds 0.9, two seeds; – = not within 12k updates; blank = not run. A: state kept across environment steps; B: no dendritic reset on a somatic spike; G: dendritic input scaled by $\sqrt{\tau_d/2}$. GRU: continuous-state reference core.

	A	B	G	τ_d	$L=8$	$L=16$	$L=32$
Spiking-WM				2	– / –	– / –	12k / –
				16	– / –	– / –	
				128	– / –	– / –	
B only		✓		128	– / –		
G only			✓	128		– / –	
A	✓			2	– / 9k	– / –	
	✓			128	3k / 4k	8k / 8.5k	
A+B	✓	✓		128	3k / 3k	4.5k / 4.5k	– / –
A+B+G	✓	✓	✓	128	**0.5k / 0.5k**	**0.5k / 0.5k**	**2k / 3k**
GRU core	continuous state				0.5k / 0.5k	1k / 1k	5k / 3.5k

Write gain. Removing both resets does not make long time constants useful by itself. The Euler update in Eq. (1) writes input with gain $1/\tau_c$, so for white input the stationary spread of v^c shrinks as $1/\sqrt{2\tau_c}$: a long time constant retains a cue but stores it faintly. With both resets removed (Sec. 3.2), a cue moved the basal potential of a $\tau_c = 128$ dendrite by 0.004–0.009, about 1% of the threshold $\vartheta = 0.5$, against 0.19 at $\tau_c = 2$.

3.2. Causal test

We test these mechanisms with interventions on Spiking-WM in a world-model probe that isolates memory from exploration and credit assignment. The data are BSuite memory-length episodes [31] collected by a random policy, with the terminal reward set to the cue (± 1, *memlen-L-recall*); only the world model is trained, and we measure the AUC of its reward prediction at the recall step on held-out episodes, from 0.5 without memory to 1 with perfect recall (details in Supp. C). Three interventions target the mechanisms above: **A** keeps all compartments across environment

steps and clears them only at episode starts; **B** keeps the dendrites when the soma fires; **G** multiplies the dendritic input by $\sqrt{\tau_d/2}$, which equalizes the stationary spread across dendritic time constants τ_d. Table 1 gives four findings.

With resets, time constants do not matter. Spiking-WM fails at 8 and 16 steps for $\tau_d \in \{2, 16, 128\}$, and the cue can be decoded from its spikes for only 2–3 steps after it is shown. The one exception supports this diagnosis: at 32 steps, one of two seeds crosses an AUC of 0.9 at the last of 12k updates by holding the cue in self-sustained recurrent firing, a memory that exists only as spikes and is learned slowly and unreliably.

Persistence is necessary. Without A, neither B nor G helps. With A and $\tau_d = 128$, the world model recalls 8 and 16 steps; removing the dendritic reset as well (A+B) nearly halves the updates needed at 16 steps (4.5k instead of 8–8.5k).

The write gain limits long time constants. A+B fails at 32 steps although a linear probe still decodes the cue from the membrane potentials at the recall step (accuracy 0.88): the trace is there, but too faint to drive spikes. Restoring the gain (A+B+G) recalls 8 and 16 steps within 500 updates and 32 steps within 2000–3000, faster than the GRU core. The gain does not work by raising the firing rate: capped at a firing rate of 0.06, A+B+G still learns 32 steps in 2000 updates.

Removing resets is not sufficient for longer memory. A+B+G fails at 64 steps even with $\tau_d = 512$, and in reinforcement learning it stops at 32 steps (Spiking-WM-NR in Fig. 1). Its time constants are counted in simulation steps, its soma still resets, and its dendrites still receive the core's spikes. Section 4 removes all three; in our model, a resetting soma alone loses 64-step recall (Sec. 5.5).

4. Fire Without Forgetting

Section 3 gives a design rule: memory must live in a state that spikes never reset, written with a gain that does not vanish for long time constants, and its recurrence should not pass through a threshold. We build FwF from neurons that follow this rule, keeping the rest of the spiking world model unchanged.

4.1. Persistent dendritic neuron

A layer has N neurons with M dendritic channels each. Its input at environment step t is the firing rate $\boldsymbol{x}_t$ of the incoming spikes over the T simulation steps, and the synaptic current into the dendrites is $\boldsymbol{w}_t = \mathrm{LN}(B\boldsymbol{x}_t) \in \mathbb{R}^{NM}$, where B is a synaptic matrix applied to spikes and LN is layer normalization [1]. Each channel has a retention time τ, in environment steps, and a decay $\lambda = e^{-1/\tau}$.

Integrator channels add the current to a leaky trace,

$$d_t = \lambda\, d_{t-1} + \sqrt{1-\lambda^2}\; w_t. \tag{3}$$

Resonant channels are damped oscillators with two state variables that rotate by a phase θ at every step:

$$\begin{pmatrix} a_t \\ b_t \end{pmatrix} = \lambda \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} a_{t-1} \\ b_{t-1} \end{pmatrix} + \sqrt{1-\lambda^2} \begin{pmatrix} w_t \\ 0 \end{pmatrix}. \tag{4}$$

An integrator channel is the special case $\theta = 0$ with $b \equiv 0$.

The soma has no state. It reads the channels of its neuron and the direct input, and emits one spike per environment step:

$$u_t = \sum_{m=1}^{M} \big(C_m a_{t,m} + C'_m b_{t,m}\big) + (D\boldsymbol{x}_t), \qquad s_t = \mathrm{H}\big(\mathrm{N}(u_t) - \vartheta\big), \tag{5}$$

where C_m, C'_m are per-neuron weights, D is a synaptic matrix, N is the normalization used throughout the world model and $\vartheta = 0.5$. The spike s_t is repeated over the T simulation steps and is the only signal the neuron sends to other neurons of the world model.

Parameters. We use $N = 512$ and $M = 4$, half of the channels of every neuron resonant. Retention times are initialized log-uniformly in $[1, 1000]$ environment steps and phases uniformly in $[0, 2\pi)$ (the initialization of the LRU [30]); both are learned. The Heaviside step is trained with a surrogate derivative, and all states are reset only at the start of an episode.

4.2. Why the neuron remembers

No reset in the memory path. The influence of a current k steps back on an integrator channel is $\partial d_t / \partial w_{t-k} = \sqrt{1-\lambda^2}\,\lambda^k$, and on a resonant channel $\sqrt{1-\lambda^2}\,\lambda^k R(k\theta)$, whatever the neuron fired in between. The memory and its gradient decay with the retention time alone and never cross a surrogate derivative on their way through time. The spike, which does cross one, is recomputed from the dendrites at every step instead of carrying the memory forward.

Normalized write. For unit-variance white input, Eq. (3) gives $\mathrm{Var}(d_t) = (1-\lambda^2)\sum_{k\geq 0}\lambda^{2k} = 1$ for every τ, whereas the Euler gain $1-\lambda$ of Eq. (1) gives $(1-\lambda)/(1+\lambda) \approx 1/(2\tau)$. The normalization of the LRU [30] thus keeps channels with retention times from 1 to 1000 steps on the same scale; intervention G is the same scaling up to a factor of 2.

What the PDN adds. We adopt the input normalization and the initialization of the LRU [30]. The PDN adds what a spiking world model needs: binary spikes as the only signal between its neurons, memory that no spike resets, a stateless soma, and separate integrator and resonant channels; Sec. 5.5

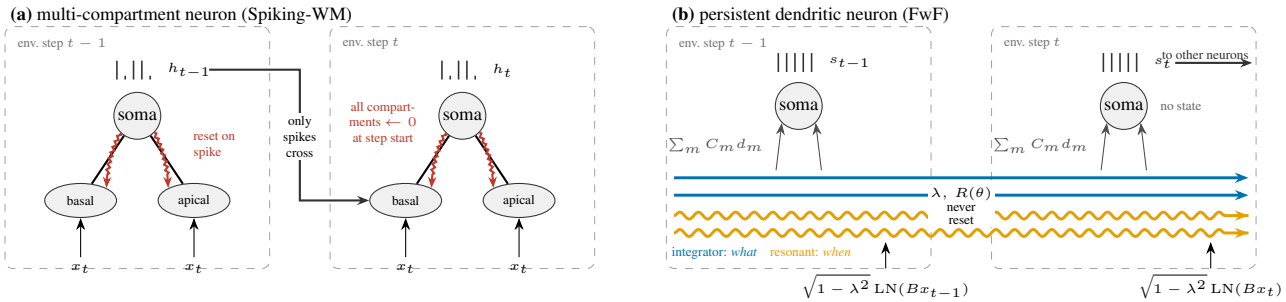


Figure 2. **Where memory lives.** (a) Spiking-WM: all compartments are cleared at every environment step and the dendrites at every somatic spike, so only the binary spikes fed back as input carry step $t-1$ to step t. (b) PDN: linear dendritic channels, never reset and without spike feedback, carry the memory across steps. Integrator channels (blue) hold a value; resonant channels (orange) rotate by a phase θ per step and so also encode elapsed time. A stateless soma reads them and emits the neuron's only output, a binary spike s_t.

removes the last three and adds spike feedback, which does not help.

A spectrum of retention times. The memory length a task needs is not known in advance. Log-uniform retention times cover lengths from one to a thousand steps in every neuron, so no time constant is tuned per task.

Resonance tells time. An input that arrived k steps ago contributes $\lambda^k(\cos k\theta,\ \sin k\theta)\, w_{t-k}$ to a resonant channel: its phase identifies the lag. In integrator channels, inputs of different ages are superposed with smoothly decaying weights λ^j, so isolating the input of exactly k steps ago from those of $k \pm 1$ steps ago requires subtracting channels with nearby retention times, an ill-conditioned readout. Integrators hold *what* was seen, resonators also record *when*.

Spikes communicate. All signals between neurons are spikes, so the synaptic operations $B\boldsymbol{x}_t$ and $D\boldsymbol{x}_t$ are accumulates. The dendritic recurrence and the readout $\sum_m C_m d_m$ are local to a neuron, and the normalizations need only the mean and variance of a layer, as in every spiking world model we compare: $O(NM)$ multiply-accumulates per step, against $O(N^2)$ for a dense recurrent matrix. Inside the world model, graded values never cross a synapse; they remain inside the neuron, as state variables of neuromorphic cores that support multi-variable neuron models [29]. Since the dendrites are linear in their input, the layer could also be trained with a parallel scan [39]; we train it step by step, like every other model in this paper.

4.3. Spiking world model

FwF replaces the recurrent core of the spiking world model (Sec. 3.1) by two PDN layers; the second layer reads the spikes of the first, and its spikes form the deterministic state h_t. Everything else is unchanged and spiking: the input layer that reads (z_{t-1}, a_{t-1}), the encoder, the posterior and prior over z_t (32 categorical variables with 32 classes), and the decoder, reward and continuation heads, all built from LIF neurons with $T = 5$ simulation steps, trained with the losses of DreamerV3 [14]. The dendritic state is carried through training sequences and through imagination rollouts, and is cleared only at episode starts.

Policy input. The actor and critic read z_t, h_t and the state of the second-layer dendrites, on every task; the Memory Maze probes read the same features. The soma sends one bit per neuron and step, whereas the dendrites hold the memory in graded form. To pass them on as spikes, each state variable is standardized over the layer and encoded by T ON and T OFF thresholds at $\pm(k - \frac{1}{2})\frac{2}{T}$, $k = 1, \ldots, T$, giving $2T$ binary frames whose rates represent $2T + 1$ levels. With six state variables per neuron, the actor and critic receive 6144 additional binary inputs per simulation step, the only path by which the dendritic state leaves a neuron; their first layer reads these inputs through a rank-64 factor. Supp. E compares a policy that reads (z_t, h_t) only.

5. Experiments

5.1. Setup

Models. All models share one spiking world model, the DreamerV3-style network of Spiking-WM [41] at the size of DreamerV3-S (512 units, $T = 5$), and differ in the recurrent core: **Spiking-WM** (multi-compartment neurons with both resets, $\tau = 2$); **recurrent LIF** neurons, cleared at every environment step as usual in spiking RL; **GRSN** [34], gated recurrent spiking neurons updated once per environment step with their state carried; **Spiking-WM-NR**, Spiking-WM with A+B+G (Sec. 3.2) and $\tau_d = 128$, *i.e.* without both resets; **DreamerV3** [14], the same architecture without spikes (GRU core, SiLU, one simulation step); and **FwF** with two PDN layers. As the state of the art among ANN world models on memory tasks we also report **R2I** [37], a

Table 2. **Memory in reinforcement learning.** BSuite memory length [31]: environment steps until the evaluation return first reaches 0.9 (– = not within the budget); †lengths, budget and windows of R2I [37] (1M steps, one seed per environment; action-indexed reward head). POPGym RepeatPrevious [25], environment settings of R2I's code: final return, mean of the last three evaluations (chance = −0.5). n/a: not run, as LIF and GRSN fail at 32 steps. ‡R2I, official code and configuration, 2M steps: mean training-episode return over the last 200k steps (for our models the two measures differ by at most 0.04). Blue: in progress.

	BSuite memory length: env. steps to return 0.9 ↓							RepeatPrevious: return ↑	
	L=8	16	32	64	51†	71†	101†	Easy	Medium
Spiking-WM [41]	–	–	–	–	–	–	–	0.43	-0.50
Recurrent LIF	62k	–	–	–	n/a	n/a	n/a	0.69	-0.51
GRSN [34]	32k	112k	–	–	n/a	n/a	n/a	0.33	-0.50
Spiking-WM-NR	12k	22k	52k	–	–	–	–	-0.29	-0.50
DreamerV3 (ANN) [14]	12k	32k	62k	–	902k	–	–	0.39	-0.44
R2I (ANN) [37]	reported in [37], see text							-0.50‡	-0.49‡
FwF (ours)	12k	22k	42k	132k	102k	222k	262k	0.68	0.57
budget (env. steps)	150k	200k	300k	500k	1M	1M	1M	2M	2M

DreamerV3 agent with a state-space recurrence, run with its official code and configuration on POPGym, and with its published results on BSuite. Within a benchmark, all models use identical hyperparameters, budgets and evaluation. The actor and critic of every model read (z_t, h_t), where h_t is the spike output of the core (for DreamerV3 its GRU state); those of FwF also read its dendritic state as spikes (Sec. 4.3), which gives its actor-critic 4.7M parameters instead of 3.4M (Supp. A); at exactly 3.4M it still learns every BSuite length and RepeatPrevious Medium (Supp. E). On POPGym, FwF has 8.6M world-model parameters, DreamerV3 7.5M and Spiking-WM 7.2M.

Benchmarks. We use benchmarks with published protocols and ANN results. *BSuite memory length* [31] shows a cue, then L uninformative steps, and rewards the agent for recalling the cue; we run $L \in \{8, 16, 32, 64\}$, and $L \in \{51, 71, 101\}$ with the lengths, budget (1M steps) and training windows of R2I [37]. *POPGym RepeatPrevious* [25] rewards the agent at every step for naming the suit of the card shown k steps earlier ($k = 4$ Easy, $k = 32$ Medium), one of the three most memory-demanding POPGym tasks that R2I evaluates; we use the environment settings of R2I's code (eight parallel environments, train ratio 64, unscaled rewards) for 2M steps. *Memory Maze 9×9* [32] is first-person 3D navigation with 1000-step episodes; we use its offline probing benchmark with the official dataset and probes: after 1M world-model updates with truncated backpropagation through time, probes decode the positions of the target objects (MSE) and the maze layout (accuracy) from the frozen state. *DMC* [43] from proprioception (walker walk, cheetah run, cartpole swingup; 500k steps, train ratio 512, as in DreamerV3) is our non-memory control. Spiking-WM was evaluated on DMC from pixels, where its 19M-parameter model reaches 938, 577 and 791 on these tasks after 1M frames [41]; here it runs inside the shared DreamerV3-S-sized world model. On BSuite the reward is a function of cue *and* action, which a scalar reward head of a Dreamer world model fails to learn within our budgets; all models therefore use a reward head indexed by the action (Supp. B). This head also helps DreamerV3, which solves 51 steps here, whereas R2I reported it failing beyond about 30.

World-model probes. To measure memory without exploration and credit assignment, we also train world models alone on random-policy data and measure the AUC of their reward prediction where reward depends on the past (Sec. 3.2): *memlen-L-recall*, where the terminal reward is the cue L steps back, and *RepeatPrevious*, where the reward indicates whether the action names the card k steps back. We report updates until the AUC exceeds 0.9 within 12k (memlen) or 10k (RepeatPrevious) updates.

Evaluation. Evaluations average 32 episodes on BSuite (every 10k steps, 20k with R2I's lengths) and 16 on POPGym (every 50k), with the mode of the policy; Supp. A reports how the return holds after it first reaches 0.9.

5.2. Memory in reinforcement learning

Table 2, Fig. 1 and Fig. 3 show that every spiking baseline stops at a memory length set by its resets. Spiking-WM never recalls a cue, not even 8 steps back. Recurrent LIF, cleared at every step, solves 8 steps; GRSN, which keeps its state but resets its membrane at every spike, solves 16; Spiking-WM-NR, without both resets, solves 32. DreamerV3 solves up to 32 steps in the short budgets and 51 steps in 902k of 1M steps with R2I's lengths, and fails at 64, 71 and 101.

FwF reaches a return of 0.9 at every length. It is as fast as the best baseline up to 16 steps, the fastest at 32 (42k steps), and the only model that reaches 0.9 at 64 steps (132k of 500k steps); after that its return is at least 0.5 in 97% of the evaluations (mean 0.91), whereas no other model

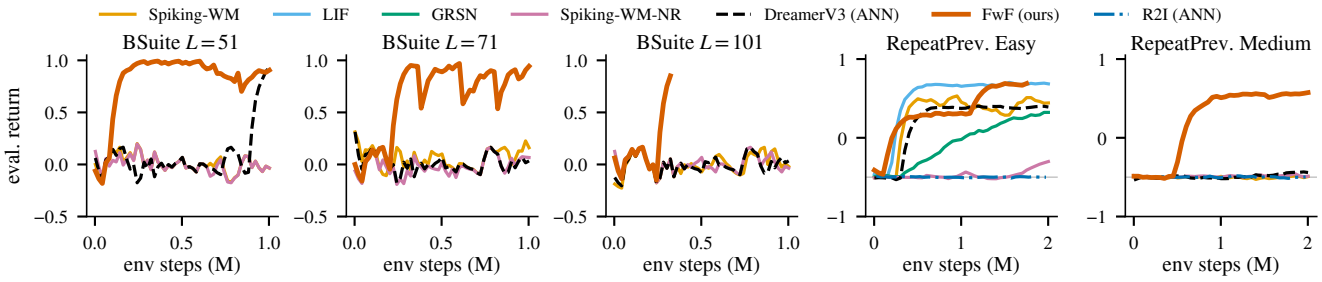


Figure 3. **Learning curves** on BSuite memory length with the lengths and budget of R2I (left three) and POPGym RepeatPrevious (right two; chance $= -0.5$). Exponential moving average of the evaluation return (R2I: training-episode return, see Tab. 2). FwF's runs are in progress.

exceeds 0.5 at any evaluation. This does not depend on the RL budget: in the world-model probes, DreamerV3 fails at 64 steps as well (Sec. 5.3). With R2I's lengths it reaches 0.9 at 51, 71 and 101 steps after 102k, 222k and 262k environment steps, with a mean return of 0.93, 0.85 and 1.00 over the evaluations that follow and final returns of 0.90, 0.96 and 1.00 (Fig. 3); DreamerV3 needs 902k steps for 51 and reaches 0.9 at neither 71 nor 101. R2I reports a median return of about 0.9 at 51 and 71 memory steps and about 0.35 at 101 after 1M steps (10 seeds; Fig. 7 of [37]), with DreamerV3's default reward head; our lengths count one step more, and all our models use the action-indexed reward head (Sec. 5.1), so the comparison is indicative. These BSuite results do not depend on the dendritic readout: with an actor and critic that read (z_t, h_t) only, FwF reaches 0.9 at every length, at 64 steps even sooner (102k), whereas on POPGym and DMC walker walk the readout is needed (Supp. E).

On RepeatPrevious Easy, FwF reaches a return of 0.68 after 88% of the budget, while the final returns of the baselines range from -0.29 (Spiking-WM-NR) to 0.69 (recurrent LIF). On Medium, all baselines remain at chance after 2M steps, DreamerV3 included, whereas the return of FwF has risen to 0.57 after 100% of the budget. R2I, run with its official code, stays at chance on both tasks within 2M steps. Its published curves span 50M steps: on Medium they leave chance only after about 3M steps, and on Easy they rise after about 1M steps and average 0–0.3 at 2M with a large spread across seeds, while DreamerV3 learns Easy but not Medium (Fig. 9 of [37]), as in our runs. At 2M steps, the comparison is one of sample efficiency. **[TODO: Rank-64 actor (switched Oct. 6): recheck the BSuite comparisons above once L=16–101 finish (Oct. 7–8); final POPGym returns (Medium Oct. 6, Easy Oct. 8).]**

5.3. What the world model remembers

The probes separate memory from policy learning (Tab. 3). On memlen-L-recall, FwF learns every length at least as fast as any other model and is the only one that recalls 64 steps (3.5k updates); DreamerV3 and Spiking-WM-NR stop at 32, GRSN at 16, LIF at 8. The same ordering as in RL shows that the limits of the baselines are limits of their world models.

Table 3. **World-model probes** (random-policy data, no policy). Updates until the reward AUC exceeds 0.9; – = not within 12k (memlen) or 10k (RepeatPrevious) updates; in parentheses the final AUC (0.5 = no memory) when it ends below 0.9. *Held out (not used during development): 20k updates, training window 128.

	memlen-L-recall				RepeatPrevious		
	L=8	16	32	64	k=4	32	64*
Spiking-WM [41]	– (0.49)	– (0.54)	12k (0.66)	– (0.52)	5k	– (0.50)	– (0.50)
Recurrent LIF	6k	– (0.52)	– (0.50)	– (0.50)	4k	– (0.50)	– (0.50)
GRSN [34]	1.5k	4k	– (0.51)	– (0.51)	– (0.89)	– (0.51)	– (0.50)
Spiking-WM-NR	0.5k	0.5k	2k	– (0.51)	– (0.81)	– (0.54)	– (0.54)
DreamerV3 (ANN) [14]	0.5k	1k	6k	– (0.50)	3k	– (0.50)	– (0.50)
FwF (ours)	0.5k	0.5k	1.5k	3.5k	2k	6k	7k

RepeatPrevious asks for a different memory: not one value held for long, but the input of an exact lag among a stream of similar inputs. At $k = 4$ most models learn it, FwF fastest (2k updates). At $k = 32$, FwF is the only model that learns it (AUC 0.93 after 6k updates); all others, including DreamerV3 and GRSN, remain at 0.50–0.55 (curves in Supp. C). Resonance is the reason: the same model with integrator channels only also stays at chance (0.54), and in RL its world model reached a reward AUC of only 0.55 on Medium (Supp. F). We added resonant channels after the integrator-only model failed this task, and the configuration has not changed since. RepeatPrevious Hard ($k = 64$), which we did not use during development, tests it on a new lag: FwF reaches an AUC of 0.9 after 7k updates, and all other models stay at 0.50–0.54 for 20k updates.

5.4. Memory Maze

[TODO: Memory Maze runs reach 1M updates around Oct. 10–13; the values shown are the latest in-training probes. Final values: frozen probes (200k probe updates) on the policy features.] Memory Maze tests whether a

Table 4. **Memory Maze and DMC.** Memory Maze 9×9 offline probing [32]: probes on the frozen state after 1M updates, second half of the evaluation episodes (constant predictor: 23.9 MSE, 80.8% walls). DMC from proprioception: return after 500k steps. Bottom: reference results from [32] (larger RSSM).

	Memory Maze 9×9		DMC return ↑		
	obj. MSE ↓	walls % ↑	walk	run	swing
Spiking-WM [41]	11.8	83.2	923	343	497
Recurrent LIF	11.7	83.6	893	478	739
GRSN [34]	11.6	83.3	820	349	628
Spiking-WM-NR	11.4	83.4	884	389	559
DreamerV3 (ANN) [14]	9.8	85.3	953	699	871
FwF (ours)	9.1	83.0	908	470	755
RSSM [32]	8.3	89.0	n/a	n/a	n/a
RSSM, TBTT [32]	6.2	94.0	n/a	n/a	n/a

Table 5. **Ablations** on the world-model probes (as in Tab. 3). Each row changes one design choice of FwF.

	memlen-L-recall		RepeatPrev.	firing
variant	L=32	64	k=32	rate
FwF	1.5k	3.5k	6k	0.18
dendrites cleared every step	n/a	– (0.50)	n/a	0.18
single $\tau = 16$	n/a	– (0.50)	n/a	0.18
single $\tau = 128$	1k	3k	n/a	0.19
τ fixed at initialization	1.5k	4k	n/a	0.17
Euler write gain $1 - \lambda$	3.5k	– (0.50)	n/a	0.17
one PDN layer	2k	– (0.51)	n/a	0.18
no resonant channels	1.5k	4k	– (0.54)	0.17
all channels resonant	1.5k	6.5k (0.86)	5k	0.17
firing rate capped at 0.06	2k	– (0.52)	n/a	0.06
soma with membrane and reset	n/a	– (0.50)	6k	0.17
spike feedback into dendrites	n/a	4k	7k	0.16

world model builds a map of a 3D maze from first-person views over 1000 steps. Table 4 reports the probes on the state that the policy reads: (z_t, h_t) for all models, and in addition the dendritic spikes for FwF. **[TODO: Text after the results: compare FwF with the spiking baselines and DreamerV3; the memory gain is the difference to a probe on the current image embedding alone (no memory).]**

5.5. Ablations

Table 5 removes one design choice at a time (curves in Supp. C). *Persistence*: clearing the dendrites at every environment step erases the memory of FwF (AUC 0.50 at 64 steps), the direct test of reset-induced amnesia inside our own model. *Retention time*: a single retention time of 16 steps fails at 64 steps and one of 128 succeeds; the log-uniform spectrum succeeds without being set for the task, learned or kept at its initialization (3.5k vs. 4k updates). *Write gain*: with the Euler gain $1 - \lambda$ the long-retention channels are too faint, and the model fails at 64 steps. *Depth*: one PDN layer fails at 64 steps; the spikes of the first layer give the second a nonlinear input. *Channel types*: without resonant channels the model fails RepeatPrevious Medium; with only resonant channels it still solves it but needs almost twice as many updates at 64 steps (6.5k vs. 3.5k, and its AUC does not stay above 0.9), where a value must be held unchanged. Half of each kind keeps both strengths. *Sparsity*: capping the firing rate at 0.06 (from 0.18) keeps 32-step recall (2k updates) but loses 64-step recall within the budget. *Stateless soma*: a soma that keeps a membrane potential across steps and is reset by its spike learns $k = 32$ as fast as FwF (6k updates) but fails at 64 steps (AUC 0.50). *Spike feedback*: dendrites that also receive the layer's spikes learn both tasks slightly more slowly (4k and 7k updates instead of 3.5k and 6k).

5.6. Non-memory control and energy

[TODO: Final DMC returns of FwF (Oct. 7) and pixel walker walk (Oct. 9–10).] On DMC (Tab. 4), FwF reaches 908, 470 and 755 on walker walk, cheetah run and cartpole swingup, against 923, 343 and 497 for Spiking-WM and 953, 699 and 871 for DreamerV3 in the same setting.

Per environment step (Supp. H), the agent of FwF, world-model filter and actor, costs 5.2 μJ, 4.3× less than DreamerV3 (22.0 μJ); the filter alone costs 3.7 against 17.2 μJ, at a recurrent firing rate of 0.16. Most of its operations are accumulates triggered by spikes, including those of the 6144 dendritic spike inputs of the actor (1.5 μJ for the whole actor); the multiply-accumulates of the dendrites are local to each neuron (3 per integrator and 6 per resonant channel and step). The spiking baselines cost 2.4–3.9 μJ but do not remember beyond 8–32 steps. These estimates count 32-bit arithmetic only [16], the convention of SNN energy reports; memory access, which a hardware-aware study shows can dominate [5], is not included for any model.

6. Conclusion

Spiking world models forget because their neurons are reset at every environment step and at every spike, which leaves their time constants nothing to act on. FwF keeps memory in persistent, normalized integrator and resonant dendrites that no spike resets, and communicates through the spikes of a stateless soma. A spiking world model built from these neurons remembers further back than DreamerV3 on BSuite and recalls exact lags that DreamerV3 cannot, at a recurrent firing rate of 0.16 and 4.3× less arithmetic energy per step.

Limitations. The dendritic state is graded, so deployment needs neuromorphic cores with multi-variable neurons and layer normalization, and our energy estimate counts arithmetic but not memory access.

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016. 4

[2] Malyaban Bal and Abhronil Sengupta. P-SpikeSSM: Harnessing probabilistic spiking state space models for long-range dependency tasks. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[3] Guillaume Bellec, Darjan Salaj, Anand Subramoney, Robert Legenstein, and Wolfgang Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 1, 2

[4] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739, 2021. 3

[5] Manon Dampfhoffer, Thomas Mesquida, Alexandre Valentian, and Lorena Anghel. Are SNNs really more energy-efficient than ANNs? an in-depth hardware-aware study. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 7(3):731–741, 2023. 8

[6] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018. 1

[7] Fei Deng, Junyeong Park, and Sungjin Ahn. Facing off world model backbones: RNNs, transformers, and S4. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1, 2

[8] Maxime Fabre, Lyubov Dudchenko, Younes Bouhadjar, and Emre Neftci. SiLIF: Structured state space model dynamics and parametrization for spiking neural networks. *arXiv preprint arXiv:2506.06374*, 2025. 3

[9] Wei Fang, Zhaofei Yu, Zhaokun Zhou, Ding Chen, Yanqi Chen, Zhengyu Ma, Timothée Masquelier, and Yonghong Tian. Parallel spiking neurons with high efficiency and ability to learn long-term dependencies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[10] David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 1, 2

[11] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019.

[12] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020. 1

[13] Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering Atari with discrete world models. In *International Conference on Learning Representations (ICLR)*, 2021.

[14] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. 1, 2, 3, 5, 6, 7, 8

[15] Saya Higuchi, Sebastian Kairat, Sander M. Bohte, and Sebastian Otte. Balanced resonate-and-fire neurons. In *International Conference on Machine Learning (ICML)*, 2024. 3

[16] Mark Horowitz. Computing's energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. 8

[17] Justus Huebotter, Pablo Lanillos, Marcel van Gerven, and Serge Thill. Spiking neural networks for continuous control via end-to-end model-based learning. *arXiv preprint arXiv:2509.05356*, 2025. 2

[18] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000. 3

[19] Eugene M. Izhikevich. Resonate-and-fire neurons. *Neural Networks*, 14(6–7):883–894, 2001. 2, 3

[20] Matthew Larkum. A cellular mechanism for cortical associations: An organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151, 2013. 3

[21] Guisong Liu, Wenjie Deng, Xiurui Xie, Li Huang, and Huajin Tang. Human-level control through directly trained deep spiking Q-networks. *IEEE Transactions on Cybernetics*, 53 (11):7187–7198, 2023. 2

[22] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. 3

[23] Chris Lu, Yannick Schroecker, Albert Gu, Emilio Parisotto, Jakob Foerster, Satinder Singh, and Feryal Behbahani. Structured state space models for in-context reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[24] Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are sample-efficient world models. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[25] Steven Morad, Ryan Kortvelesy, Matteo Bettini, Stephan Liwicki, and Amanda Prorok. POPGym: Benchmarking partially observable reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 6

[26] Steven Morad, Ryan Kortvelesy, Stephan Liwicki, and Amanda Prorok. Reinforcement learning with fast and forgetful memory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[27] Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. 3

[28] Tianwei Ni, Michel Ma, Benjamin Eysenbach, and Pierre-Luc Bacon. When do transformers shine in RL? decoupling memory from credit assignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[29] Garrick Orchard, E. Paxon Frady, Daniel Ben Dayan Rubin, Sophia Sanborn, Sumit Bam Shrestha, Friedrich T. Sommer, and Mike Davies. Efficient neuromorphic signal processing with Loihi 2. In *IEEE Workshop on Signal Processing Systems (SiPS)*, pages 254–259, 2021. 1, 2, 5

[30] Antonio Orvieto, Samuel L. Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De. Resurrecting recurrent neural networks for long sequences. In *International Conference on Machine Learning (ICML)*, 2023. 2, 4

[31] Ian Osband, Yotam Doron, Matteo Hessel, John Aslanides, Eren Sezener, Andre Saraiva, Katrina McKinney, Tor Lattimore, Csaba Szepesvári, Satinder Singh, Benjamin Van Roy, Richard Sutton, David Silver, and Hado van Hasselt. Behaviour suite for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2020. 1, 2, 3, 6

[32] Jurgis Pasukonis, Timothy Lillicrap, and Danijar Hafner. Evaluating long-term memory in 3D mazes. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 6, 8

[33] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37 (6):989–999, 2003. 3

[34] Lang Qin, Ziming Wang, Runhao Jiang, Rui Yan, and Huajin Tang. GRSN: Gated recurrent spiking neurons for POMDPs and MARL. In *AAAI Conference on Artificial Intelligence*, pages 1483–1491, 2025. 2, 5, 6, 7, 8

[35] Jan Robine, Marc Höftmann, Tobias Uelwer, and Stefan Harmeling. Transformer-based world models are happy with 100k interactions. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[36] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575:607–617, 2019. 1

[37] Mohammad Reza Samsami, Artem Zholus, Janarthanan Rajendran, and Sarath Chandar. Mastering memory tasks with world models. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2, 5, 6, 7

[38] Shuaijie Shen, Chao Wang, Renzhuo Huang, Yan Zhong, Qinghai Guo, Zhichao Lu, Jianguo Zhang, and Luziwei Leng. SpikingSSMs: Learning long sequences with sparse and parallel spiking state space models. In *AAAI Conference on Artificial Intelligence*, pages 20380–20388, 2025. 3

[39] Jimmy T. H. Smith, Andrew Warrington, and Scott W. Linderman. Simplified state space layers for sequence modeling. In *International Conference on Learning Representations (ICLR)*, 2023. 5

[40] Matei-Ioan Stan and Oliver Rhodes. Learning long sequences in spiking neural networks. *Scientific Reports*, 14:21957, 2024. 3

[41] Yinqian Sun, Feifei Zhao, Mingyang Lyu, and Yi Zeng. Spiking world model with multicompartment neurons for model-based reinforcement learning. *Proceedings of the National Academy of Sciences*, 122(50):e2513319122, 2025. 1, 2, 3, 5, 6, 7, 8

[42] Guangzhi Tang, Neelesh Kumar, Raymond Yoo, and Konstantinos P. Michmizos. Deep reinforcement learning with population-coded spiking neural network for continuous control. In *Conference on Robot Learning (CoRL)*, 2020. 2

[43] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018. 6

[44] Aaron Voelker, Ivana Kajić, and Chris Eliasmith. Legendre memory units: Continuous-time representation in recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2

[45] Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari S. Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[46] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in Neuroscience*, 12:331, 2018. 3

[47] Bojian Yin, Federico Corradi, and Sander M. Bohté. Accurate and efficient time-domain classification with adaptive spiking recurrent neural networks. *Nature Machine Intelligence*, 3 (10):905–913, 2021. 2

[48] Dehao Zhang, Malu Zhang, Shuai Wang, Jingya Wang, Wenjie Wei, Zeyu Ma, Guoqing Wang, Yang Yang, and Haizhou Li. Dendritic resonate-and-fire neuron for effective and efficient long sequence modeling. *arXiv preprint arXiv:2509.17186*, 2025. 3

[49] Shimin Zhang, Qu Yang, Chenxiang Ma, Jibin Wu, Haizhou Li, and Kay Chen Tan. TC-LIF: A two-compartment spiking neuron model for long-term sequential modelling. In *AAAI Conference on Artificial Intelligence*, pages 16838–16847, 2024. 1, 2

[50] Weipu Zhang, Gang Wang, Jian Sun, Yetian Yuan, and Gao Huang. STORM: Efficient stochastic transformer based world models for reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[51] Hanle Zheng, Zhong Zheng, Rui Hu, Bo Xiao, Yujie Wu, Fangwen Yu, Xue Liu, Guoqi Li, and Lei Deng. Temporal dendritic heterogeneity incorporated with spiking neural networks for learning multi-timescale dynamics. *Nature Communications*, 15:277, 2024. 1, 3